

12.421 / 12.621 Lecture 8

Scattering from the ocean and other
surfaces w/ small scale roughness

Topics covered

- Ocean wave spectra
- Scattering from disttly rough surfaces (Bragg scattering)
- Scattering from undulating, slightly rough surfaces

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• Overview

- We will use scattering of EM waves from the ocean surface as an illustrative example of scattering from a broad range of natural surfaces

→ Advantage of doing so,

- Allows us to incorporate concepts from the beginning of the semester on dielectric properties because temperature and salinity have important effects on scattering from the ocean

- Nicely illustrates the influence of surface roughness at varying wavelengths and sources

- To get started, let us consider the geometry of the surface and the processes that influence it

• Ocean surface

- Elevation of the ocean surface is influenced by a variety of factors

- Ocean surf (cont.)

- Roughly in order of decreasing wavelength

- Gravitational equipotential surface (geoid),
which approximates mean sea level and
is the elevation that the ocean would take
given a planet's mass distribution and
rotation, and in the absence of
external forcing (tides, winds, etc.)

- Tides

- Atmospheric pressure gradients

- Currents

- Wind-driven waves → of greatest interest
in today's lecture

- Wind-driven surface waves

- Ocean surface waves are mechanical waves
that exist at the ocean-atmosphere
boundary

- As with any mechanical wave, they are defined
by a perturbing force and a restoring force

- In all cases of interest in this lecture, the
perturbing forces are shear stresses caused
by the wind

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Ocean waves (cont.)

- We will focus our attention on two types of waves defined by different restoring forces

1. Gravity waves are long wavelength ($> 2 \text{ cm}$) whose restoring force is gravity.

- Swells, which build up from wind forcing over long timescales are examples of gravity waves
- Can play an important role in scattering from the ocean surface
- Terrestrial analogs: rolling hills, crop rows

2. Capillary waves are short wavelength ($< 2 \text{ cm}$)

whose restoring force is surface tension

- Produce small-scale roughness that is superimposed on longer-wavelength roughness
- Play a leading role in EM scatter from ocean surface
- Terrestrial analog is any small-scale roughness: rocks, small erosional features
- At radar wavelengths, it is often the case that the portion of the wave spectrum of greatest interest lies at intermediate wavelengths

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• Ocean waves (cont.)

- At these intermediate wavelengths, both gravity and surface tension act to restore the ocean surface $\rightarrow$ such waves are called gravity-capillary waves
- Dispersion relation for gravity-capillary waves in deep water (depth $\gg$ wavelength) is given as

$$\omega^2 = gK + \frac{\gamma}{\rho_w} K^3$$

where

$g \equiv$ gravitational acceleration

$\gamma \equiv$ surface tension

$\rho_w \equiv$ mass density of water

- In the limits

- small K : $\omega \approx \sqrt{gK}$ for gravity waves

- large K : $\omega \approx \sqrt{\frac{\gamma K^3}{\rho}}$

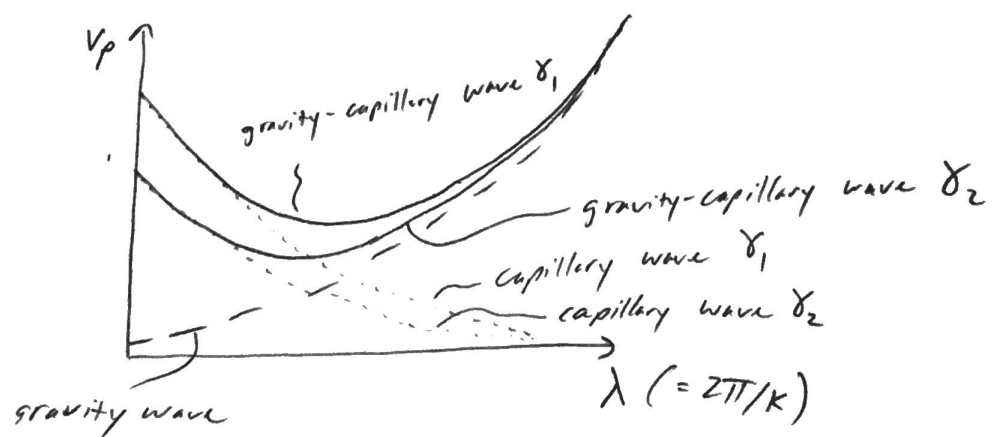
- We can see from this expression how "surfactants" (substances that reduce surface tension) change the dispersion relation of ocean surface waves $\Rightarrow$ change the scattering properties of EM waves

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- Common examples of surfactants include mineral oil (as in the oil spill example that we have seen many times so far in this course) and biological films. Microplastics may also act as a surfactant, but I believe this remains an open question.
- Gravity-capillary waves are excited (grow) in a turbulent wind field and reach equilibrium when the wind speed matches the phase velocity of the surface waves. From the dispersion relation, we derive the phase velocity

$$v_p = \frac{\omega}{k} = \sqrt{\frac{g}{k} + \frac{\gamma}{\rho} k}$$

- In general, surfactants will reduce the surface tension of clean water $\Rightarrow$ (all else being equal) will slow (gravity-)capillary waves



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- When wave speed (phase velocity) matches wind speed, the RMS height h of the ocean can be defined as

$$h^2 = \int_0^\infty \Lambda(\omega) d\omega$$

where the frequency spectrum Λ is related to local wind speed u_w through empirical relation

$$\Lambda(\omega) = \frac{g^2}{10\omega^5} \exp\left\{-\frac{3}{4}\left(\frac{g}{u_w\omega}\right)^4\right\}$$

$\Rightarrow$ strong correlation between wind speed and surface height, which, of course, we all know from experience

- For remote sensing, we are concerned with the direction of the winds relative to the look direction (or plane of incidence of EM wave) of the sensor. This relation is usually given as an empirical relation between the normalized radar cross section σ_{rp}^0 and the angle α between plane of incidence and wind direction

$$\sigma_{rp}^0 = A u_w^n (1 + a \cos \alpha + b \cos 2\alpha)$$

where A, a, b, n are all inferred from observations. This inference requires data collected from multiple look directions to provide info on wind speed and direction

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- Simple models of EM scatter from the ocean (and other randomly rough surfaces)

- Let's dig a bit more into the mechanisms of EM scattering from the ocean surface

- We'll start with the most useful model for surface scattering (at least at microwave frequencies):

Bragg scattering or small-perturbation model

- Applicable when RMS height of the surface and the horizontal length scale of surface roughness are small compared w/ EM wavelength; canonically

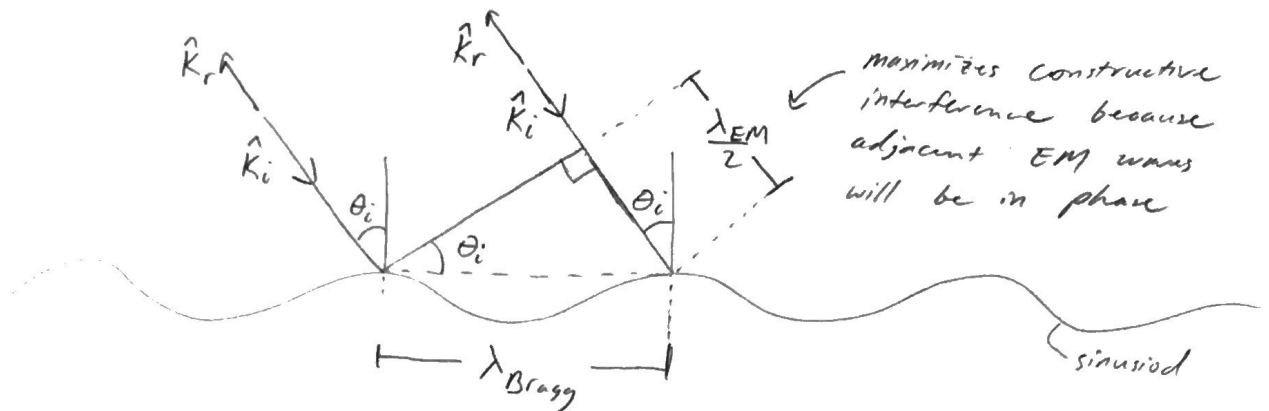
$$Kh < 0.3 \quad \text{and} \quad \frac{h}{L} < 0.3$$

- In general, h corresponds only to the component of the roughness spectrum responsible for the majority of scattering $\rightarrow$ called the Bragg wavelength

- Basic idea of the model: Any random surface can be decomposed into a series of sinusoids. The wavelength that contributes most to the backscatter power is one that maximizes constructive interference (resonance) of scattered waves

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Bragg scattering



From geometry, we find

$$\lambda_{Bragg} = \frac{\lambda_{EM}}{2 \sin \theta_i}$$

or

$$K_{Bragg} = 2 K_{EM} \sin \theta_i = \frac{2\pi}{\lambda_{Bragg}} \equiv \text{Bragg wave number}$$

- To apply the concept of Bragg scattering to backscattered power, Rice (1951) applied perturbation methods and derived an expression for the first-order (meaning, higher-order terms in the Taylor Series expansion are dropped) normalized radar cross section

$$\sigma_{pp}^0 = 8 K_{EM}^4 h^2 \cos^4 \theta_i |\alpha_{pp}|^2 W(K_{Bragg}, 0)$$

where

$W(K_x, K_y) \equiv$ surface roughness spectrum discussed in the previous lecture

$$\left(W = \frac{l^2}{2} \exp \left\{ - (l K_{EM} \sin \theta_i)^2 \right\} \text{ for Gaussian surface} \right)$$

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• Bragg scattering (cont.)

- The scattering coefficients are

$\alpha_{hh} = \rho_h \equiv$ Fresnel reflection coef for h polarized waves at oblique incidence

$$\alpha_{vv} = - \left[\rho_v + \tilde{\epsilon}_v^2 \frac{(\epsilon_2 - 1) \tan^2 \theta_i}{2\epsilon_2} \right]$$

$$= (\epsilon_2 - 1) \frac{\sin^2 \theta_i - \epsilon_2 (1 + \sin^2 \theta_i)}{(\epsilon_2 \cos \theta_i + \sqrt{\epsilon_2 - \sin^2 \theta_i})^2}$$

where ρ_v and $\tilde{\epsilon}_v$ are the Fresnel reflection and transmission coeffs for v-pol at oblique incidence

and we have assumed $\epsilon_1 \approx 1$

- Why does $\alpha_{hh} = \rho_h$ but $\alpha_{vv} \neq \rho_v$?

A. Goes back to EM boundary conditions: Recall from Faraday's Law we got the condition that the tangential component of the electric field is continuous across the surface; h field is entirely tangent to the surface for all roughness (at least to a reasonable approximation); whereas tangential component of v field changes w/ roughness

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Brass scattering (cont.)

- Key take-aways

1. $\sigma_{hv}^0 = \sigma_{vh}^0 = 0$ to first order (higher-order factors introduce small nonzero values of σ_{hv}^0 (and σ_{vh}^0) but these are generally negligible as they are in the noise of RS instruments)
2. σ_{pp} represents total scattered power and is
 - a. polarization-dependent
 - b. function of dielectric properties and viewing geometry (indirectly EM wavelength)
 - c. independent of characteristics of surface roughness
3. W influences the direction of scattering and is
 - a. dependent on characteristics of roughness
 - b. dependent on EM wavelength and viewing geometry
 - c. independent of dielectric properties
 - d. independent of polarization
4. $\sigma_{vv}^0 > \sigma_{hh}^0$ due to the expression for reflectivity (thus, due to EM BCs)

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• Tilted Bragg scatter model

- Previous model only applies to horizontal surfaces w/
small-scale roughness
- Can combine the Bragg model w/ Kirchhoff model
for undulating surfaces by (again) treating
the surface as a bunch of randomly rough
surfaces (infinite extent) that are tilted
due to longer-wavelength roughness features
- On the ocean, this simply means that we represent
the wind-driven gravity-capillary waves with a
Bragg scatter model and rotate our local
coordinate system to account for the passage
of gravity waves (swells)
- This is a very accurate, simple model of scattering
from the ocean surface for incidence angles
ranging from $30^\circ - 65^\circ$. (Lower incidence angles
produce primarily specular scatter while angles
 $> 65^\circ$ involve different mechanism that we
won't discuss in this course because they are
esoteric)

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• Tilted Bragg (cont.)

- If we define the tilting of the rough surface patches in terms of two angles

$\psi \equiv$ in-plane (plane of incidence) tilt

$\zeta \equiv$ anti-plane (" " ") tilt

it can be shown (Valenzuela, 1978) the normalized radar cross section is

$$\sigma_{pg}^0 = 8 K_{EM}^4 h^2 \cos^4 \theta_i \Gamma_{pg} W$$

where

$$W = W(2 K_{EM} \sin(\theta_i + \psi), 2 K_{EM} \cos(\theta_i + \psi) \sin \zeta)$$

and the polarization-dependent reflectivities are

$$\Gamma_{pp} = \left| \left(\frac{\sin(\theta_i + \psi) \cos \zeta}{\sin \theta_m} \right)^2 \alpha_{pp} + \left(\frac{\sin \zeta}{\sin \theta_m} \right)^2 \alpha_{gg} \right|^2$$

for $p \neq g$

$$\Gamma_{hv} = \Gamma_{vh} = \left(\frac{\sin(\theta_i + \psi) \sin \zeta \cos \zeta}{\sin \theta_m} \right)^2 |\alpha_{vv} - \alpha_{hh}|^2$$

with

$$\theta_m = \cos^{-1}(\cos(\theta_i + \psi) \cos \zeta) \equiv \text{incidence angle}$$

measured relative to the surface normal

and

α_{hh} and α_{vv} as defined in the Bragg model

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• Tilted Bragg model

- This model reduces to the previous (untilted) model when $\psi = \xi = 0$

- Take aways:

- Tilting modifies both reflectivity and effective roughness spectrum

- But dependencies of reflectivity and roughness spectrum remain the same as in Bragg model and also include tilt angles

- Cross-polarized scattering (σ_{hv}^0) is nonzero when anti-plane tilt angle $|\xi| > 0$

→ because in this case, neither the h-polarized nor v-polarized electric fields are locally tangent to surface $\Rightarrow$ conservation of tangential component of E field results in a change of polarization

- And these are the basic concepts of EM scattering from surfaces